TIME CONSISTENCY AND THE THE CREDIBILITY OF MONETARY MONETARY POLICY

An exploration of why central banks struggle to maintain credible low-inflation policies and what can be done about it.

STRUCTURE

TODAY'S AGENDA

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INTRODUCTION

The time inconsistency problem in monetary policy

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THE MODEL

Central bank preferences and constraints

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Single interaction between CB and public

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Trust equilibria and self-fulfilling expectations

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POLICY IMPLICATIONS

Solutions to the credibility problem

CHAPTER 1

INTRODUCTION

WHAT IS TIME INCONSISTENCY?

THE TIME INCONSISTENCY PROBLEM

A policy is optimal when announced but becomes suboptimal when implementation time arrives. The policy aims to induce a desirable outcome, but if that outcome fails to materialise, following through becomes pointless.

CLASSIC EXAMPLE: FLOOD PLAIN HOUSING

1

ANNOUNCEMENT

Government declares no relief for flood plain construction

2

DETERRENT EFFECT

Policy appears optimal to prevent risky building

3

REALITY CHECK

Houses built anyway; flood occurs

4

POLICY REVERSAL

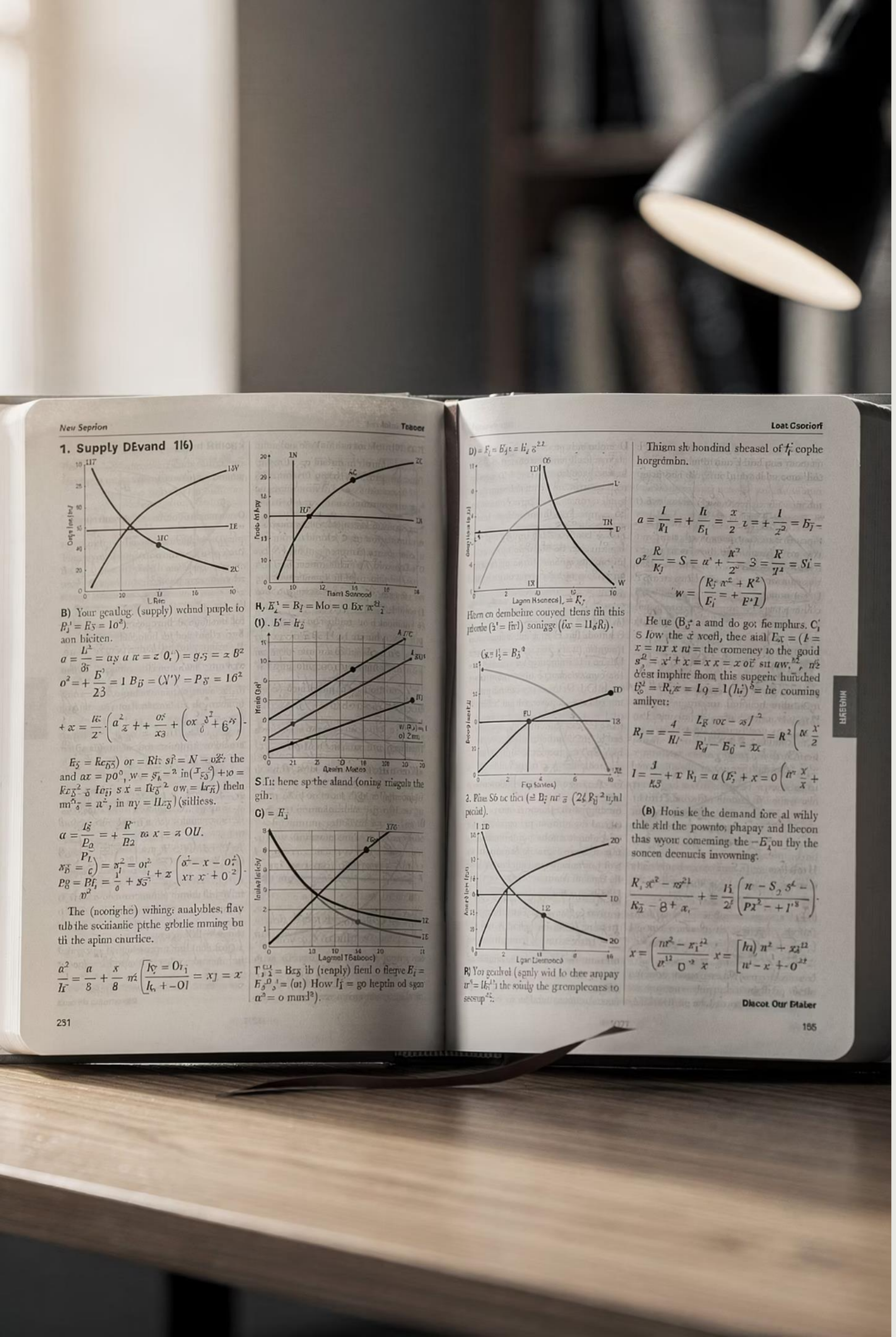
Public pressure forces government to provide relief

NOBEL PRIZE WINNERS

Finn Kydland and Ed Prescott first identified the time inconsistency problem in monetary policy, earning the 2004 Nobel Prize in Economics.

KEY INSIGHT

Even well-intentioned policymakers face systematic incentives to renege on announced policies once the public has formed expectations.



MONETARIST AND NEW CLASSICAL FOUNDATIONS

LONG-RUN NEUTRALITY

Monetary policy cannot affect real output in the long run

SHORT-RUN EFFECTS

Output responds only to unpredictable monetary surprises

POLICY IMPLICATION

Central banks should focus on low inflation, not output goals

THE TEMPTATION TO INFLATE

Suppose the central bank announces zero inflation and the public believes it, setting expected inflation to zero.

The CB can then 'fool' the public by expanding money supply faster than expected, creating positive inflation.

From the Lucas aggregate supply equation, output rises when actual inflation exceeds expected inflation.

The Core Problem: A zero inflation policy is *ex ante* optimal but ceases to be so if the public actually believes it will be implemented.

CHAPTER 2

THE MODEL

BUILDING THE FRAMEWORK

THREE KEY ASSUMPTIONS

ASSUMPTION 1

The central bank directly controls inflation (simplified but justified by Friedman's super-neutrality arguments)

ASSUMPTION 2

Economic agents form expectations rationally

ASSUMPTION 3

The CB dislikes inflation and deviations from target GDP

THE CENTRAL BANK LOSS FUNCTION

The CB seeks to minimise:

$$L_t = \frac{a}{2}(\pi_t)^2 + \frac{b}{2}(Y_t - Y^T)^2$$

INFLATION COMPONENT

Coefficient *a* reflects aversion to inflation

OUTPUT COMPONENT

Coefficient *b* reflects preference for hitting GDP target YT

POLITICAL BIAS AND POLICY PREFERENCES

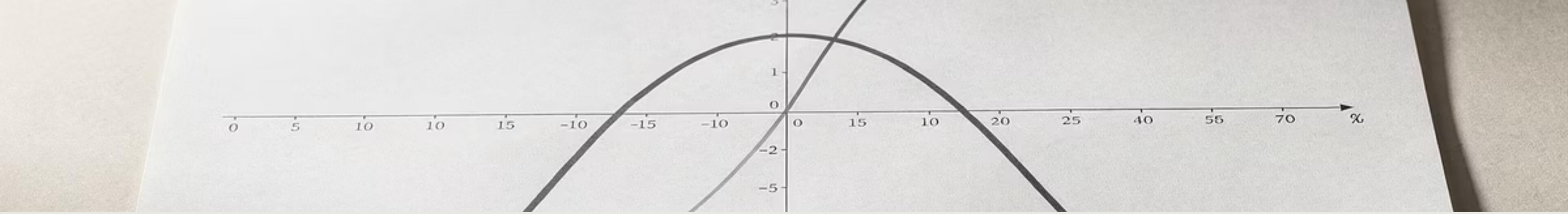
Define the CB's political bias as $\lambda \equiv YT - Y^*$

LEFT-LEANING CB

- High λ ($YT > Y^*$)
- Targets unemployment below natural rate
- More interventionist

HAWKS VS DOVES

- High b/a ratio: inflation dove
- Low b/a ratio: inflation hawk
- Ratio determines policy stance



QUADRATIC LOSS FUNCTION PROPERTIES

The loss function is quadratic in both inflation and output deviations.

Key Implication: Marginal disutility of deviations increases with the size of the deviation. Small misses are tolerable; large misses are severely penalised.

INCORPORATING THE LUCAS AS EQUATION

The Lucas aggregate supply equation (modified for inflation):

$$Y_t - Y = \kappa(\pi_t - E_{t-1}\pi_t)$$

This can be rewritten to incorporate the CB's bias:

$$Y_t - Y^T = \kappa(\pi_t - E_{t-1}\pi_t) - \lambda$$

Output deviates from target when inflation surprises occur or when the CB has a political bias.

DERIVING THE COMPLETE LOSS FUNCTION

Substituting the Lucas AS equation into the loss function yields:

$$L_t = \frac{a}{2}(\pi_t)^2 + \frac{b}{2}\kappa^2(\pi_t - E_{t-1}\pi_t)^2 - b\lambda\kappa(\pi_t - E_{t-1}\pi_t) + \frac{b}{2}\lambda^2$$

This formulation captures the trade-offs the CB faces between inflation control and output stabilisation.

CHAPTER 3

THE ONE-SHOT GAME

SINGLE INTERACTION ANALYSIS

SEQUENCE OF EVENTS

1

PERIOD -1: ANNOUNCEMENT

CB announces inflation rate A-1π

2

PERIOD -1: EXPECTATIONS

Public forms expectation E-1π

3

PERIOD 0: IMPLEMENTATION

CB implements actual inflation π

4

TIME CONSISTENCY?

Check if π = A-1π

RATIONAL EXPECTATIONS CONSTRAINT

With rational expectations, any systematic CB policy will be perfectly anticipated by the public.

Therefore, Y cannot deviate from Y^* through monetary policy (unless policy is random).

Ex Ante Optimal Policy: Accept that $Y - YT = -\lambda$ and minimise inflation loss by setting $\pi = 0$.

TESTING ZERO INFLATION CREDIBILITY

Suppose the CB announces A-1π = 0 and the public believes it, setting E-1π = 0.

When implementation time arrives, the CB faces:

$$L = \frac{a}{2}\pi^2 + \frac{b}{2}\kappa^2\pi^2 - b\kappa\lambda\pi + \frac{b}{2}\lambda^2$$

Taking the first-order condition:

$$\pi = \frac{b\kappa\lambda}{a + b\kappa^2} > 0$$

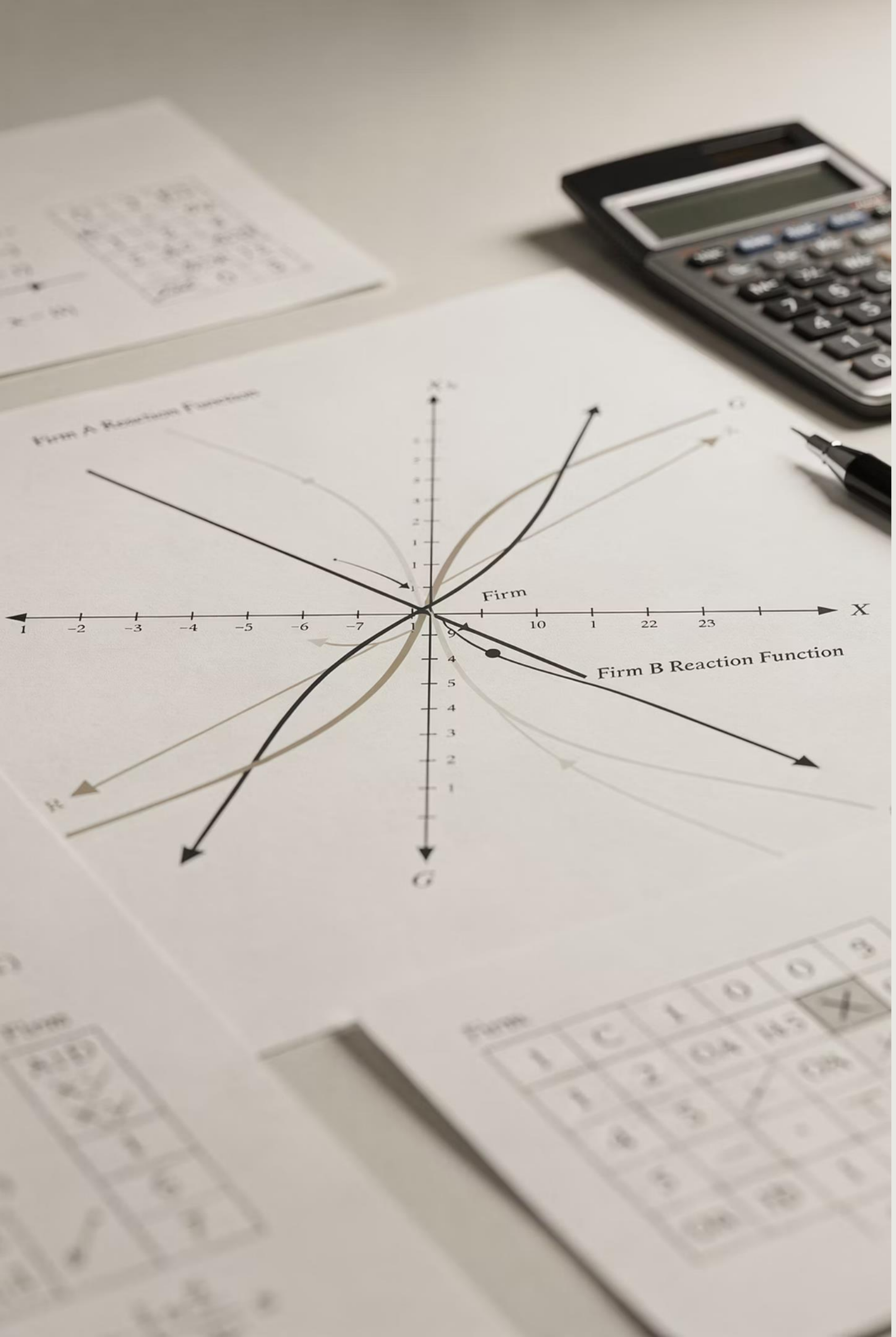
TIME INCONSISTENCY ESTABLISHED

$\Pi \neq A\text{-}1\Pi$

Zero inflation announcements are time-inconsistent. The CB has a systematic incentive to renege.

With rational expectations, agents will not believe non-credibly low inflation announcements.

The Credibility Problem: Rational agents will not believe announcements promising unreasonably low inflation.



SOLVING BACKWARDS FOR EQUILIBRIUM

To find the time-consistent policy, solve backwards. Taking A-1π and E-1π as given, the CB's optimal inflation is:

$$\pi = \frac{bk\lambda}{a + bk^2} + \frac{bk^2}{a + bk^2} E_{-1}\pi$$

This is the CB's **reaction function**: for any public expectation, the CB's optimal response.

EQUILIBRIUM CONDITION

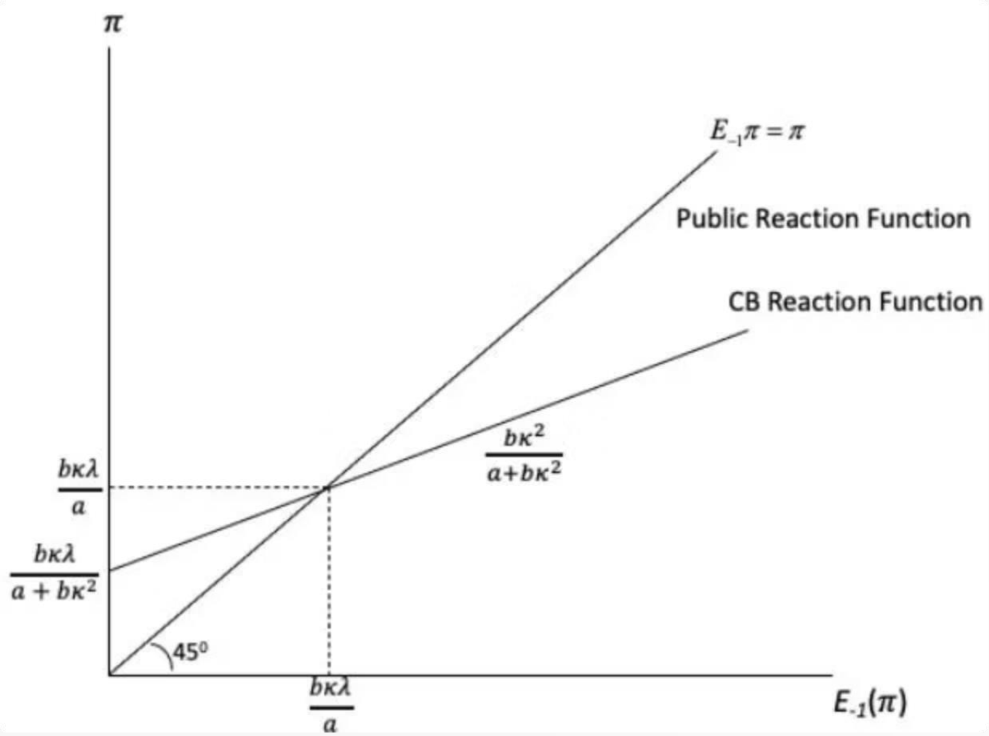
For equilibrium, $\pi = E^{-1}\pi$ (the public's reaction function).

Solving:

$$\pi = \frac{b\kappa\lambda}{a}$$

Both the public's rational expectation and the CB's announcement equal this value.

> This is the unique time-consistent equilibrium of the one-shot game.



REACTION FUNCTIONS DIAGRAM

The intersection of the CB reaction function and the public's 45° line determines the equilibrium inflation rate bkλ/a.

CHAPTER 4

COMPARING CB LOSSES

FOUR STRATEGIC OUTCOMES

LOSS UNDER HONEST ZERO INFLATION

If the CB announces zero inflation, is believed, and delivers it:

$$L = \frac{b\lambda^2}{2}$$

This loss arises purely from the output gap (Y* - YT), with no inflation penalty.

LOSS UNDER EQUILIBRIUM POLICY

Under the time-consistent equilibrium, output remains at Y* but inflation is positive:

$$L^e = \frac{b\lambda^2}{2}\left[\frac{b\kappa^2}{a} + 1\right]$$

This is strictly greater than $\mathcal{L}^*$. The CB suffers both the output gap loss and an inflation penalty.

LOSS FROM CHEATING

If the public expects zero inflation but the CB implements $\pi = b\kappa\lambda/(a + b\kappa^2)$, output rises to:

$$Y = Y + \frac{b\kappa^2\lambda}{a + b\kappa^2}$$

The overall loss from cheating:

$$L^c = \frac{a}{a + b\kappa^2}\frac{b}{2}\lambda^2 < \frac{b}{2}\lambda^2$$

Cheating yields lower loss than honesty, explaining the temptation to renege.

LOSS FROM THE PURIST STRATEGY

If the CB sincerely implements zero inflation but the public expects $bk\lambda/a$, a recession occurs:

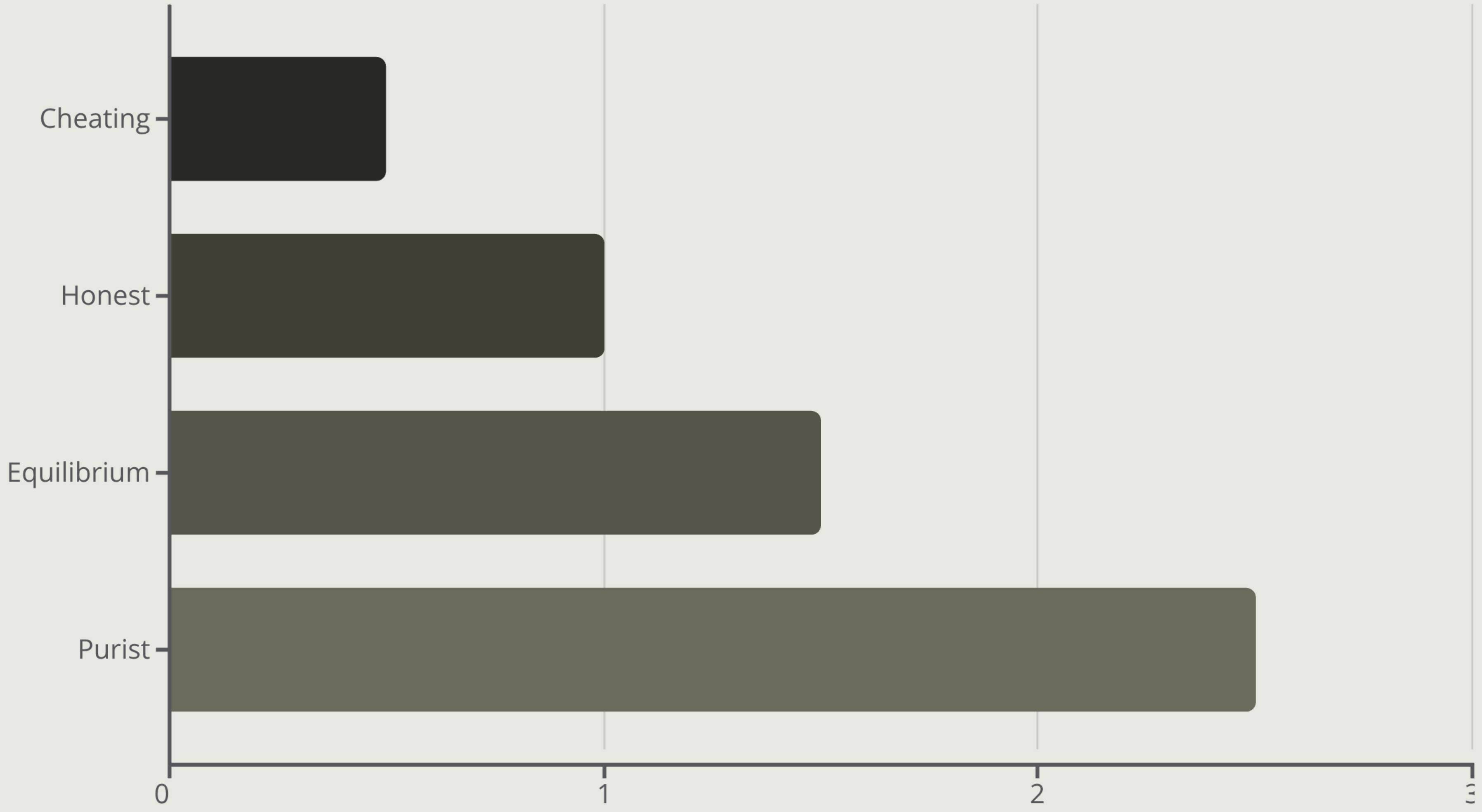
$$Y - Y^T = -[1 + \frac{bk^2}{a}]\lambda$$

The purist loss:

$$L^p = \frac{b}{2}\lambda^2 [1 + \frac{bk^2}{a}]^2 > L^e$$

Insisting on zero inflation when not believed produces the worst outcome.

SUMMARY OF LOSSES



The credibility problem locks the CB into an unnecessarily high inflationary stance. More liberal CBs (high λ, high b/a) face higher time-consistent inflation.

CHAPTER 5

REPEATED INTERACTION

TRUST AND REPUTATION

INFINITE HORIZON GAME STRUCTURE

The CB and public interact repeatedly with no end date. Each period follows the same sequence as the one-shot game.

The CB's overall loss function:

$$L = L_0 + \beta L_1 + \beta^2 L_2 + \cdots$$

where $\beta < 1$ is the discount factor (future losses matter less than present ones).

THE ROLE OF REPUTATION

After period 0, the public bases future expectations on past CB behaviour.

INITIAL TRUST

Public believes initial announcement

FRAGILE CREDIBILITY

If CB cheats once, trust is lost forever

PERMANENT PUNISHMENT

Future expectations lock onto time-consistent rate $b\kappa\lambda/a$

THE INTERTEMPORAL TRADE-OFF

CHEATING PAYOFF

- One-period gain: $\mathcal{L}^* - \mathcal{L}c$
- Temporary reduction in loss

CHEATING COST

- Permanent future excess loss: $\mathcal{L}e - \mathcal{L}^*$
- Discounted by β each period

The CB trades off temporary gain against permanently higher discounted future losses.

CONDITIONS FOR TRUST EQUILIBRIUM

If the discount factor β is sufficiently high, the present value of future excess losses outweighs the one-period gain from cheating.

Trust Equilibrium: Zero inflation can emerge as an equilibrium when the CB values the future highly enough (high β).

MULTIPLE EQUILIBRIA PROBLEM

Trust equilibrium is not the only possibility. The public might distrust the CB from the outset.

TRUST EQUILIBRIUM

Public believes announcements; CB delivers low inflation

DISTRUST EQUILIBRIUM

Public disbelieves announcements; CB trapped at high inflation

SELF-FULFILLING

Both expectations are rational and confirmed by outcomes

SELF-FULFILLING PROPHECIES

When multiple equilibria exist, initial expectations can be self-fulfilling if enough agents hold them.

This provides theoretical support for Keynes' idea of self-fufilling prophecies in rational expectations models.

KEY RESULTS SUMMARY

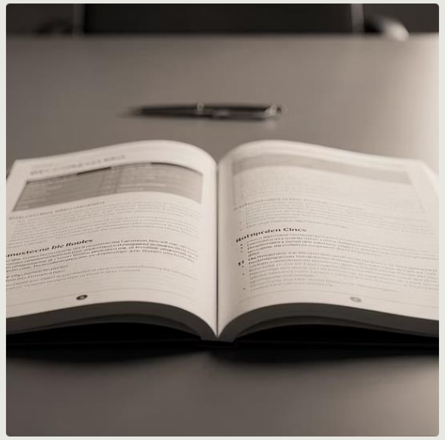
1. Under rational expectations, systematic policy cannot raise Y_t above Y^*
2. But temptation to 'fool' the public always exists
3. Very low inflation announcements lack credibility
4. Only credible inflation rates can be implemented ($b\kappa\lambda/a$ in one-shot game)
5. Infinite repetition enables zero inflation, but multiple equilibria exist

CHAPTER 6

POLICY IMPLICATIONS

SOLUTIONS TO CREDIBILITY

FOUR POLICY SOLUTIONS



1. RULES VS DISCRETION

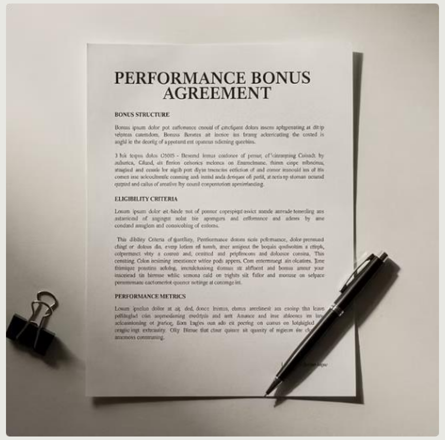
Implement formula-based policy (e.g., Taylor Rule) to eliminate discretion and cheating ability

2. CONSERVATIVE BANKER

Appoint inflation hawk with high *a*, low *b*, low λ (e.g., Paul Volcker 1979)

3. FIXED EXCHANGE RATE

Import credibility from low-inflation country by pegging currency



4. PERFORMANCE PAY

Link central banker compensation to inflation target achievement

CONCLUSION

The time inconsistency problem reveals fundamental tensions in monetary policymaking. Credibility cannot be assumed—it must be built through institutional design.

4

POLICY SOLUTIONS SOLUTIONS

Institutional mechanisms to enhance credibility

2

EQUILIBRIA

Trust and distrust both self-fulfilling

1

CORE INSIGHT

Announcements alone insufficient without commitment